ScienceDirect

Expert Systems with Applications

Date: 1 October 2022

Show more

Research article Get rights and content

A majority–minority cellular automata algorithm for global optimization

Juan Carlos Seck-Tuoh-Mora, Norberto Hernandez-Romero, Fredy Santander-Baños, Valeria Volpi-Leon, Joselito Medina-Marin, Pedro Lagos-Eulogio Show more

Cite Add to Mendeley Share 10.1016/j.eswa.2022.117379

Access through your organi... Purchase PDF

Article preview Recommended articles Cited by (9) Metrics

Highlights

- Majority and minority CA are useful to aim an algorithm for global optimization.
- The new algorithm (MmCAA) is very simple and easy to implement.
- MmCAA is inspired in the neighborhood concept and concurrent rules of CA.
- MmCAA is competitive to optimize test problems and engineering applications.

Article preview

Abstract

Introduction

Section snippets

References (59)

Abstract

Cellular automata (CA) are discrete dynamical systems that can give rise to complex behaviors under certain conditions. Its operation is based on simple local interactions between its elements. The different dynamical behaviors of CA offer a great diversity of ideas and inspiration to propose new metaheuristics focused on global optimization. One such automata is the one specified by the majority rule, which is capable of implementing logical operations under the right conditions. Taking this rule as inspiration, this work proposes the majority–minority CA algorithm. This algorithm takes different adaptations of the majority rule and its counterpart, the minority rule, to establish different rules that modify vectors of real values in order to achieve a good balance in exploration and exploitation tasks for optimization tasks. The efficiency of the majority–minority CA algorithm is tested with 50 widely used test problems in the literature, using both uni- and multimodals and fixed dimensions. Additionally, 3 engineering applications used in recent literature are also optimized. The numerical results verify the competitiveness of the algorithm compared to other recently published specialized algorithms. The source codes of the proposed algorithm are publicly available at https://github.com/juanseck/MmCAA.git.

Keywords

Global optimization; Majority cellular automata; Metaheuristics; Engineering applications

Previous article in this issue

Next article in this issue

Introduction

The optimization of engineering problems represents an increasingly complex task due to the significant number of components and interactions between them. Conventional algorithms face difficulties with these types of problems, such as convergence to a local optimum, stagnation and dependence on the correct selection of initial conditions to converge appropriately. Additionally, for their correct operation, the traditional methods based on mathematical programming and the application of the conjugate gradient and quasi-Newton methods require the problem to be optimized to be compliant with specific mathematical properties, such as convexity, continuity and differentiability, which often implies oversimplifying the problem to be addressed(Gogna and Tayal, 2013, Sarker et al., 2003). This issue puts them at a disadvantage in solving problems characterized as being multi-dimensional, multimodal, non-continuous and non-differentiable(Jamil & Yang, 2013). Thus, traditional optimization methods no longer offer quick solutions and satisfactory results.

This situation has led to the search and definition of new optimization methods such as metaheuristics, which can be described as algorithms that adaptively and intelligently combine various procedures to solve complex problems. Although in most cases, it is not possible to show that metaheuristics reach the optimal value of a problem, they are capable of producing good results in a satisfactory amount of time(Kumar & Davim, 2020).

Metaheuristics are also characterized by their easy specification and simple operation that does not depend on the gradient of the functions to be optimized, thus facilitating their computational implementation. It is desired that a metaheuristic can escape the local minima in multi-dimensional problems and be applied in a wide range of problems for the optimization, design and identification of the parameters of engineering systems.

Among the proposed metaheuristics are the swarm intelligence (SI) algorithms, where a set of agents interact locally and without global control to share and generate new information through a simple set of rules that generates collective behavior, emerging with the ability to optimize complex problems(Hassanien & Emary, 2018).

Most SI algorithms are inspired by the behaviors that arise between members of a natural system. Generally, its operation begins with a population that evolves toward values close to the optimum of a problem. Additionally to showing flexibility and adaptability in a great variety of cases, this type of SI algorithm has the advantage of not depending on the selection of initial conditions and not requiring special mathematical conditions of the problem(Cui and Gao, 2012, Slowik, 2021).

In contrast, the SI algorithm presented in this work is inspired by the behavior of an artificial system known as a cellular automaton, in particular, to use the adaptations of a special type of rule known as the majority rule and its complement, the minority rule(Goles et al., 2018, Martinez et al., 2012). This article aims to present a new optimization method called the majority–minority cellular automata algorithm (MmCAA), which proposes several adaptations of the majority and minority rule for global optimization tasks for problems defined in multiple dimensions. Another objective is to compare the MmCAA with other recent metaheuristics recognized for their efficiency and robustness over a set of test functions with varying degrees of difficulty. Finally, the application of the MmCAA in various engineering problems will be demonstrated to prove its efficiency and versatility in real cases.

An SI algorithm must perform in an efficient and balanced manner two main actions: exploration and exploitation. As the operations for each type of action can be counterproductive and leave the algorithm trapped in a local minimum, an adequate strategy that allows the algorithm to adapt during the application of its different operations, calculate reasonable solutions, avoid getting stuck in the search space and have a premature convergence toward the local minima must be implemented.

The MmCAA algorithm works by first defining a random set of initial solutions. Each of these solutions will generate a neighborhood of new possible solutions, applying adaptations of the majority and minority rule to promote changes to the current solutions that, in some cases, take only the information contained in each solution. In others, information between solutions is communicated to create new solutions. New solutions will be selected based on how much they improve the fitness (or cost) in the problem to be optimized. In the case of this work, the minimization of functions.

The strength of the proposed algorithm is that its specification and implementation are uncomplicated, facilitating its application in optimization tasks and resulting in the achievement of satisfactory results compared with other state-of-the-art metaheuristics.

The contribution of the MmCAA consists of showing that specific cases of cellular automata (CA) that have been studied and applied for other purposes due to their complexity can also be adapted to propose new metaheuristics designed for the optimization of multimodal and non-differentiable problems in multiple dimensions.

As metaheuristics have also been used in other fields of engineering, such as the identification of parameters in complex models, with multiple parameters to be tuned(Kumar & Davim, 2020), the operation of the MmCAA is also tested in various engineering problems. Specifically, the MmCAA is proved for the step-cone pulley design(Rao, 1996, Savsani and Savsani, 2016), the multi-plate disc clutch brake design(Wang et al., 2020) and the speed reducer design(Savsani & Savsani, 2016), comparing the results with other recognized metaheuristics and obtaining competitive results.

The structure of the paper is as follows. Section2 presents a review of the literature of the latest metaheuristics proposed for the global optimization of functions, emphasizing the SI metaheuristics. Section3 introduces the basic concepts and main work of the CA with the majority and minority rule. Section4 explains the MmCAA proposal and the different adaptations of the majority and minority rule, to carry out exploration and exploitation actions concurrently and the general strategy of the MmCAA. Section5 describes the experimental results obtained by taking 50 test

functions in 30 and 500 dimensions. A statistical study with 13 other state-of-the-art algorithms is also presented. Section6 applies the MmCAA in 3 engineering problems that are taken from specialized literature. These results are also compared with reports from other recent algorithms to show the effectiveness of the MmCAA. The last section provides the conclusions and future proposals for developing new algorithms based on CA.

Organizational access

Get full-text access by signing in with your organisation

Access through **your organization**

Other access options

Purchase PDF

Need help with access? ↗

Section snippets

SI metaheuristics review

Metaheuristics based on SI refer to algorithms that mimic the collective intelligence emerging from the global behavior of systems composed of multiple agents that follow simple interaction rules without the existence of any central control. Systems with multiple agents have been an inspiration for the generation of SI algorithms that seek the best set of values to optimize a given objective function with a simple implementation.

SI-based metaheuristic algorithms began to gain popularity in the …

Basics of majority rule CA

CAs are discrete dynamical systems proposed by Von Neumann and Ulam to show the feasibility of defining self-reproducing systems that work with uncomplicated local interactions of their components(von Neumann, 1966). The simplicity of the specifications of a CA makes it ideal for implementation in a computer. Its theoretical study and applications have been developed over the years to characterize its ability to produce complex behaviors(McIntosh, 2009, Wolfram, 2002).

A CA comprises cells …

MmCAA

The aim of the MmCAA is to emulate the dynamic behavior of the majority and minority rules in CA. Each cell (called a smart-cell) is the set of values defining a solution for the problem to be optimized. The change of each smart-cell is defined by a set of rules, where one is chosen randomly to update its position. Different rules

generate different positions, from which the new position with the best cost in the function to be optimized is selected, replacing the position of the smart-cell. …

Experimentation with benchmark functions

Table1 presents the 50 test functions used in this work to prove the effectiveness of the MmCAA.

In this set of benchmark functions, the first 16 unimodal functions assess the proposed algorithm's exploitability. The following 17 multimodal test functions show the exploration and escape properties of the local minima. Finally, the remaining 17 functions are fixed-dimension test functions and various local minimums to test the balance of exploration and exploitation movements carried out by the …

Step-cone pulley design (SCP)

This design problem consists of optimizing the design of a 4 step-cone pulley that has a minimum weight(Rao, 1996, Savsani and Savsani, 2016). In this problem, we have 5 design variables, 4 of which correspond to the diameters of the pulleys ($y(1)$, $y(2)$, $y(3)$ and $y(4)$) at each step and the last design variable is the pulley width ($y(5)$). This design is subject to 11 restrictions; three of them are equalities, and the others are inequalities, which ensures that the same length of the belt is …

Conclusions and future work

This work presents a new algorithm for global optimization based on the majority and minority rules of CA called MmCAA. This algorithm also applies concepts inspired by CA such as local interaction and neighborhoods between smart-cells. The randomness and concurrency of applying the different rules generate an adequate balance between exploration and exploitation actions as well as the exchange of information between solutions during the optimization process.

Algorithm tuning and benchmarking …

CRediT authorship contribution statement

Juan Carlos Seck-Tuoh-Mora: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Norberto Hernandez-Romero:** Conceptualization, Methodology, Validation, Writing – original draft. **Fredy Santander-Baños:** Investigation, Methodology, Writing – original draft. **Valeria Volpi-Leon:** Investigation, Methodology, Writing – original draft. **Joselito Medina-Marin:** Conceptualization, Formal analysis, Methodology. **Pedro Lagos-Eulogio:** Conceptualization, Validation, …

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

Acknowledgments

This study was supported by the National Council for Science and Technology (CONACYT), Mexico with project number CB- 2017-2018-A1-S-43008 and FOP16-2021-01-320109. Fredy Santander-Baños was supported by CONACYT, Mexico grant number 639377. …

References (59)

AbualigahL. *et al.*
The arithmetic optimization algorithm
Computer Methods in Applied Mechanics and Engineering (2021)

AbualigahL. *et al.*
Aquila optimizer: A novel meta-heuristic optimization algorithm
Computers & Industrial Engineering (2021)

BraikM.S.
Chameleon Swarm Algorithm: A bio-inspired optimizer for solving engineering design problems
Expert Systems with Applications (2021)

ChenH. *et al.*
An efficient double adaptive random spare reinforced whale optimization algorithm
Expert Systems with Applications (2020)

FaramarziA. *et al.*
Marine predators algorithm: A nature-inspired metaheuristic
Expert Systems with Applications (2020)

GandomiA.H. *et al.*
Krill herd: A new bio-inspired optimization algorithm
Communications in Nonlinear Science and Numerical Simulation (2012)

GolesE. *et al.*
The complexity of the asynchronous prediction of the majority automata
Information and Computation (2020)

GolesE. *et al.*
On the complexity of two-dimensional signed majority cellular automata
Journal of Computer and System Sciences (2018)

GuptaS. *et al.*
A modified sine cosine algorithm with novel transition parameter and mutation operator for global optimization
Expert Systems with Applications (2020)

HeidariA.A. *et al.*
Harris hawks optimization: Algorithm and applications
Future Generation Computer Systems (2019)

View more references

View full text

Recommended articles

Based on reading popularity

Chapter

Chapter 6 - Fabrication technologies

Patrizio Candeloro, ..., Gerardo Perozziello

Microfluidics for Cellular Applications • 2023 • Pages 223-274

Research article

Graphene-based Field Effect Diode

Abolfazl Sotoudeh, Mina Amirmazlaghani

Superlattices and Microstructures • Volume 120 • 2018 • Pages 828-836

Research article

A hybrid Grasshopper Optimization Algorithm and Harris Hawks Optimizer for Combined Heat and Power Economic Dispatch problem

Murugan Ramachandran, ..., Christober Asir Rajan Charles Gnanakkan

Engineering Applications of Artificial Intelligence • Volume 111 • 2022 • Article 104753

Research article

Deep long short term memory based minimum variance kernel random vector functional link network for epileptic EEG signal classification

Sebamai Parija, ..., Mrutyunjaya Sahani

Engineering Applications of Artificial Intelligence • Volume 105 • 2021 • Article 104426

Research article

An improved group teaching optimization algorithm based on local search and chaotic map for feature selection in high-dimensional data

Hamed Khosravi, ..., Vahide Babaiyan

Expert Systems with Applications • Volume 204 • 2022 • Article 117493

Research article

TOD-CNN: An effective convolutional neural network for tiny object detection in sperm videos

Shuojia Zou, ..., Marcin Grzegorzek

Computers in Biology and Medicine • Volume 146 • 2022 • Article 105543

Cited by (9)

Improved optimal foraging algorithm for global optimization ↗

Ding C., Zhu G.

Computing • Volume 106 • 2024

Improvement in Sizing Constrained Analog IC via Ts-CPD Algorithm ↗

Lagos-Eulogio P., ..., Hernández-Romero N.

Computation • Volume 11 • 2023 • Article 230

Improving Wild Horse Optimizer: Integrating Multistrategy for Robust Performance across Multiple Engineering Problems and Evaluation Benchmarks ↗

Chen L., ..., Feng C.

Mathematics • Volume 11 • 2023 • Article 3861

SLOTSA: A Multi-Strategy Improved Tunicate Swarm Algorithm for Engineering Constrained Optimization Problems ↗

Wang W., ..., Tian J.

Proceedings - 2023 IEEE International Conference on Software Services Engineering, SSE 2023 • 2023

A Multi-Disturbance Marine Predator Algorithm Based on Oppositional Learning and Compound Mutation ↗

Chen L., ..., Ma Y.

Electronics (Switzerland) • Volume 11 • 2022 • Article 4087

A New Algorithm Inspired on Reversible Elementary Cellular Automata for Global Optimization ↗

Seck-Tuoh-Mora J.C., ..., Volpi-León V.

IEEE Access • Volume 10 • 2022

View all citing articles on Scopus ↗

Metrics

Citations

Citation Indexes	9

Captures

Mendeley Readers	4

PLUMX View details ↗

© 2022 Elsevier Ltd. All rights reserved.

ELSEVIER

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

RELX™